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Compute versus accuracy tradeoff in LSTM and transformers for medical video classification

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Introduction

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Medical video classification, such as analyzing endoscopy, ultrasound, or surgical videos, plays a key role in diagnosis and treatment planning.

Deep learning models like LSTMs and Transformers are widely used for this task because they can learn patterns across video frames over time. However, these models differ greatly in computational cost and accuracy.

LSTMs process frames **sequentially**, making them memory-efficient but sometimes weaker at capturing long-range dependencies.

Transformers use **self-attention** to model relationships across all frames at once, often improving accuracy but at a higher computational cost.

Problem Statement

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This Research Project explores the tradeoff between compute requirements and classification accuracy for

Long-Short Term Memory & **Transformer Architecture**

aiming to help researchers and practitioners select suitable models for real-world medical video analysis.

Project Intent + Boundaries

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Compare LSTMs with Transformer in a classification task

- Currently for only one type of videos (already curated from an existing pipeline)
- Benchmarking will be done on a CPU only machine
- The system would have a null hypothesis which would be based of existing research, the whole research would be related around this.
- If time permits, a hybrid model shall be submitted which brings the best output

Project Timeline

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Oct 5 — Architecture Finalisation

Finalize the architecture and the design on how to approach the problem.

Oct 12 — Data Preprocessing

Clean the existing data to the best extent possible to prepare it for training.

Oct 19 — Training

Train on a CPU only Machine, while maintaining consistency and get possible accuracy.

Oct 26 — Benchmarking

Benchmarking on various fronts for both the approaches to find an optimal one.

Nov 2 — Final Analysis

Curate everything and fix any minors bugs before final presentation

Nov 9 — Paper^

^A buffer week, if time permits - hybrid model + paper.

Role of each Team Member + Status

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Adhithi M	Research + Data Processing
Anagha M Nair	Research + Architecture
Apoorv Gadiya	Training + Benchmarking
Satyam Goyal	Analysis + Data Processing

The roles are overlapping throughout the project duration but status quo right now is everyone is working on **Research** and **Literature**.

Research Papers – 1

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No.	Paper	Model relevance	Author Details and Paper
1	**Artificial Intelligence-Based Classification of Multiple Gastrointestinal Diseases Using Endoscopy Videos for Clinical Diagnosis**	ResNet + LSTM; reports spatiotemporal classification performance and model size	PubMed Central NIH
2	**Pulmonary COVID-19: Learning Spatiotemporal Features Combining CNN and LSTM Networks for Lung Ultrasound Video Classification**	CNN–LSTM; reports 93% average accuracy and 97% COVID-19 sensitivity	PubMed Central NIH
3	**Towards Accurate Surgical Workflow Recognition with Convolutional Networks and Transformers**	Directly evaluates convolutional, Transformer, and hybrid designs; reports 93% frame-level accuracy for a hybrid model	Taylor & Francis open access

Research Papers – 2

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No.	Paper	Model relevance	Author Details and Paper
4	**Video Violence Detection Using LSTM and Transformer**	Direct LSTM-versus-Transformer comparison on video violence datasets; reports AUC, speed, and model behavior	IIETA ORG
5	**Memory vs. Attention: Investigating LSTM and Transformer Models in Human Action Recognition**	Direct comparison of LSTM, BiLSTM, and Transformer classifiers for skeleton-based action recognition	IBIMA ORG
6	**CNNs, RNNs and Transformers in Human Action Recognition**	Reviews and evaluates CNN, recurrent, and Transformer approaches for human-action video recognition	arXiv full text

Research Papers – 3

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No.	Paper	Model relevance	Author Details and Paper
7	**OperA: Attention-Regularized Transformers for Surgical Phase Recognition**	Transformer temporal classification for laparoscopic video	MICCAI
8	**Temporal-Based Swin Transformer Network for Workflow Recognition of Surgical Video**	Hybrid Transformer/LSTM temporal architecture for video classification	Springer Article
9	**A Hybrid Transformer-LSTM Model with 3D Separable Convolution for Video Prediction**	Combines Transformer and LSTM temporal processing while reducing model size and computation	Santa Clara University
10	**The Discussion About Suitable Scenarios for BiLSTM and Transformer**	Reports a direct efficiency comparison, including FPS, FLOPs, and training time	ISEEE Net